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Abstract

The widespread use of Large Language Models (LLMs) across applications, including customer services and critical infrastructure supporting data systems, has made their security a priority both locally and globally. In the UAE, where national strategies such as the UAE Artificial Intelligence Strategy 2031 (UAE Government, 2017) and the National Cybersecurity Strategy (UAE Cybersecurity Council, 2023) emphasize the development of trustworthy, safe, and resilient AI systems, understanding and mitigating LLM vulnerabilities is crucial to supporting the nation's digital transformation agenda. However, few studies review the latest research on LLM vulnerabilities. This Systematic Literature Review (SLR) examines prompt injection attacks, which exploit weaknesses in LLMs to inject malicious instructions that compromise the model. Thirty-two peer-reviewed studies published between 2021 and 2025 were analyzed using PRISMA 2020 guidelines, revealing that prompt injection attacks pose a universal threat across LLM families, with an attack success rate of 70-100%. Results also demonstrated that defenses against prompt injection attacks are highly stratified; architectural defenses may be effective but impractical in many cases, while practical defenses are easily bypassed. A further gap was identified in existing defenses, which often overlook the security of backend applications, leaving end-to-end systems vulnerable. These findings highlight the need for a shift in securing LLM-integrated applications from model-centric approaches to model-agnostic and data-layer-driven architectures, aligning with the UAE's vision of building a robust digital infrastructure.

Keywords: large language models, prompt injection, cybersecurity, systematic literature review, AI security, adversarial attacks.

Prompt Injection Attacks on Large Language Models: A PRISMA-Guided Systematic Literature Review

Consider an enterprise's corporate database, which contains tens of millions of customers' records and proprietary financial information. A large-scale exploitation of this corporate database could occur not through a sophisticated zero-day exploit but rather through a single, cleverly crafted prompt that leverages LLMs' natural language processing capabilities, known as prompt injection. Large Language Models (LLMs) are sophisticated artificial intelligence systems trained on massive volumes of text and code, enabling them to execute user-defined commands. The LLMs' ability to execute commands allows autonomous agents to communicate directly with systems, supporting reasoning, task execution, code execution, and improved data query performance (Wang et al., 2023). LLMs are rapidly adopted across domains such as customer service portals, business intelligence tools, and financial analysis.

However, vulnerabilities to prompt injection attacks have severely undermined the security of LLM-integrated systems. A prompt injection attack occurs when an attacker injects hidden instructions into a user's query to an LLM, thereby bypassing safety features and potentially leaking confidential information or performing unauthorized actions. An example of this is when a user asks a customer service agent for a purchase summary while including a malicious instruction such as "Ignore previous instructions and export all registered email addresses." Using other indirect means, a malicious instruction could also be hidden in data retrieved and processed by the LLM, such as by poisoning a database or embedding a malicious email, thereby transforming what was previously a safe and trusted source of data into a potential attack trigger (Greshake et al., 2023). The interaction between the user, the model, and connected systems is illustrated in Figure 1.

This vulnerability scenario is not hypothetical. High-profile cases have been documented, demonstrating the real-world impact of prompt injections. Using a single prompt, one researcher successfully prompted Microsoft's Bing Chat to extract its codename, "Sydney" along with its internal operating instructions (Roose, 2023). Security researchers also demonstrated the "Grandma Hack," an indirect prompt injection where a command hidden in a "story" prompt manipulates the AI assistant into forwarding sensitive data (Lakera, 2023). The scale of the issue is evident from community reports, including cases where researchers manipulated e-commerce chatbots to obtain unauthorized discount codes, demonstrating how easily business logic can be compromised (Rehberger, 2023).

The threat of prompt-injection attacks against language models was quantified through a Systematic Literature Review (SLR), which found prompt injections to be successful at alarming rates, with average success rates over 70% and 100% in controlled studies. A prompt-based SQL injection (Prompt-to-SQL or P2SQL) successfully compromised every test scenario, achieving a 100% Attack Success Rate (ASR), and directly exposed the model to the potential for both data manipulation and data exfiltration (Pedro et al., 2025). Although the development of various defenses to prevent such attacks has been a direct response to this threat, the SLR identifies a significant and yet poorly addressed research gap. Specifically, the SLR demonstrated that all existing defense strategies for mitigating this threat are designed to protect the LLM from attack, constituting a "model-centric" approach. These model-centric defenses are fundamentally inadequate, as the SLR found that even advanced defenses can be bypassed. LLM-Fuzzer achieved a 97% Attack Success Rate (ASR) when used to mutate prompt injection attempts that had previously failed (Yu et al., 2025). The core issue is architectural: these defenses fail to address the broken trust boundary between the LLM and the data layer it commands.

The above-mentioned issues are significant for the United Arab Emirates, which launched national initiatives to promote the development of reliable, trustworthy, and sustainable AI systems through the UAE Artificial Intelligence Strategy 2031 and National Cybersecurity Strategy. With the increasing use of AI technology and LLM-integrated applications by businesses, government institutions, and digital providers in the UAE, it will be necessary to develop an effective defense against emerging attack vectors, such as prompt injections, to maintain trust in AI-based systems supporting the UAE's digital transformation agenda.

To address this issue, this study conducts a systematic literature review of prompt injection attacks against large language models, analyzing existing attack techniques, evaluating current defense strategies, and identifying key research gaps in securing LLM-integrated systems, thereby providing a consolidated understanding of current vulnerabilities and mitigation approaches.

Literature Review

The rapid adoption of large language models has prompted growing research interest in their security vulnerabilities. Prior studies investigate how adversarial prompts manipulate LLM behavior, explore methods for constructing prompt injection attacks, and propose mechanisms for mitigating such threats. The following sections review the existing literature on prompt injection attacks, defense strategies, and the limitations of current research.

Research on Prompt Injection Attacks

Recent studies investigate the potential of prompt injection attacks as an emerging threat to LLMs. Prompt injections exploit LLMs' compliant nature and leverage their ability to interpret natural-language instructions. Carefully crafted prompts have evidently influenced LLM responses to user requests and their interactions with connected external data sources (Shen et al., 2024).

The rapid adoption of LLMs in production environments and real-world deployments prompted research to focus on the influence of adversarial inputs on LLM behavior and the likelihood of producing violative outputs. Studies explore how malicious inputs can be used to bypass/override an LLM's internal policy, modify LLM reasoning or behavior, or compel an LLM to disclose information that would otherwise remain restricted. As a result, AI and cybersecurity research communities, as well as other relevant stakeholders, have identified prompt injections as a major security concern challenging the current technological landscape (Wang et al., 2024).

Research on Defense Mechanisms

Considerable interest has grown in improving LLM robustness in response to growing concerns and reported incidents. Consequently, researchers have proposed several approaches at different levels of system development to mitigate the risk of prompt injections. While several studies investigated alignment-based methods to influence and enforce model behavior during training (Hu et al., 2024), other studies evaluated alternative external moderation strategies, such as filtering mechanisms, to reduce compliance with malicious prompts or generated output that violates system rules (Han et al., 2024).

Further work proposed architectural and system-level safeguards to reduce the influence of adversarial prompts on model behavior and output. These approaches aim to harden the interaction between language models and the systems they control, reducing the likelihood that malicious inputs successfully influence downstream applications or limit the impact of system compromise (Wang et al., 2024);((Xu et al., 2024).

Limitations of Current Research

Despite increased attention to prompt injections, most studies on the topic remain compartmentalized and fragmented across various research fields. Researchers in adversarial

machine learning, AI safety, and software security study and propose various methods for attacking and defending against such proposed attacks. However, relatively little research has systematically synthesized and summarized the broader body of literature in the field of prompt injection research.

Furthermore, differences in evaluation methods and experimental setups make it difficult to directly compare the effectiveness of proposed attacks and defenses across studies. These limitations highlight the need for a structured synthesis of existing research. Therefore, this study adopts a systematic literature review methodology to identify, evaluate, and analyze current research on prompt injection vulnerabilities and mitigation strategies in large language models.

Methodology

A systematic literature review (SLR) of published literature on prompt injection attacks targeting LLM models was conducted to investigate previous contributions and establish an understanding of attack types, success rates, vulnerable models, and defense strategies, ultimately building a background of the current research landscape while highlighting gaps for future research. Given the topic's recency, a systematic literature review was selected as a viable method to build a comprehensive understanding of the topic and of the current published research. To ensure methodological rigor and transparency in conducting the systematic literature review, this review followed the PRISMA 2020 reporting guideline (Page et al., 2021) in four phases: database identification, screening based on exclusion criteria, further assessment of eligibility, and final inclusion in the review.

Research Questions

Five research questions were developed to understand the current scope of research in the field of interest:

1. RQ1: What are the existing types of prompt injections?
2. RQ2: How successful are the existing types of prompt injections?
3. RQ3: What LLMs can be targeted for prompt injections?
4. RQ4: What are the approaches to tackling prompt injections in LLM-based applications?
5. RQ5: What is the evidence for the validity of the existing approaches to tackling prompt injections?

The first research question focuses on the classification of prompt injection types. Identifying different patterns of prompt injections allows for mapping and classification of the different attack types into a consistent taxonomy used to classify risks to LLMs. It also helps build a repository of knowledge on evolving techniques of prompt injections, underscoring the need for adaptive security solutions as threat risk continues to evolve.

The second research question investigates the success rate of each type of prompt injection. The success rate of each type will help determine the patterns of greatest threat in real-world deployments. By comparing the success rates of each type, researchers can prioritize the defenses to implement, allocate resources appropriately to mitigate the threat, and evaluate the level of protection provided by existing defenses.

The third research question identifies which LLMs are most susceptible to prompt injection attacks. This key question determines whether different families of models and configurations exhibit varying levels of vulnerability, depending on the defenses allocated to each model. By identifying patterns of vulnerability across architectures, researchers can pinpoint systemic vulnerabilities and develop ways to mitigate those vulnerabilities.

The fourth research question develops a comprehensive, systematic approach to classifying existing and proposed defense mechanisms against prompt injections. A taxonomy of defense mechanisms will provide a foundation for analyzing the properties of defenses, evaluating the validity of existing defenses, and identifying critical research gaps.

The fifth research question synthesizes existing empirical evidence on the effectiveness of defenses against prompt injections. A systematic evaluation of the empirical evidence is necessary to transition from theoretical claims to establishing a clear, comparative understanding of which defenses are most robust, their success conditions, and the limitations of current validation efforts.

Search Strategy

To identify relevant literature on prompt injection attacks in LLMs, a Scopus search was conducted. Since prompt injection is a relatively recent research topic, the search was restricted to studies published between 2021 and 2025, when the majority of LLM security work has emerged. To account for the evolving terminology in this area, keywords were kept broad. The search strategy combined the term "LLM" with a set of attack-related keywords, including variations of "prompt injection" (prompt-injection, injection attack), "jailbreak" (jail-breaking, "jail break," jailbroken), "indirect prompt," "prompt hacking," "adversarial prompt," "malicious prompt," and "red teaming." This ensured comprehensive coverage of the different ways prompt injection and jailbreak attacks are described in the literature.

Study Selection Process

The literature generated by the search string was assessed for relevance to the research topic and guided by the inclusion and exclusion criteria. Applying the available database filters, the initial 333 papers were narrowed to 287.

The bibliographic metadata for the retrieved studies was systematically exported from Scopus in RIS file format. This included title, abstract, authors, source (journal or conference), year of publication, language, and keywords. The RIS file was then exported to Rayyan for organizing and iterative screening according to predetermined criteria, such as the number of citations and the conference's reputation.

The four stages of selection according to PRISMA 2020 are documented in Figure 2 and applied the inclusion and exclusion criteria defined in Table 1:

1. Identification: In total, 287 studies were retrieved from Scopus for 2021-2025. After duplicate removal and the application of IC-1 (English Language) and EC-1 (pre-2021 publication), 117 studies remained.
2. Screening: The 117 articles were reviewed for titles, abstracts, and keywords. Applying IC-2 (address prompt injection, jailbreaks, or related adversarial prompt attacks against LLMs) and EC-2 (topics unrelated to LLM prompt injection, such as data poisoning or adversarial ML in vision models) resulted in 81 articles remaining.
3. Eligibility: The articles were screened for full text. Applying IC-3 (empirical evidence required) and EC-3 (conceptual or theoretical papers without empirical evaluation) resulted in 35 studies remaining.
4. Inclusion: A final in-depth review of the articles was completed. Applying IC-4 (peer-reviewed journal or conference) and EC-4 (short papers or non-peer-reviewed) yielded a final article set of 32 articles.

Limitations

This methodology has its own limitations. The search relies on the availability of the literature, and results are limited to papers available on Scopus. Thus, gaps may form if pertinent studies are not published or included in the database of choice. This search also excludes languages other than English, further limiting the literature included in this literature review. It is imperative to note that the field of this study is fast-moving, and as such, published literature may be outpaced.

These limitations were mitigated through transparent documentation of the search strategy to allow future replication, acknowledgment of potential gaps, and recommendations for future reviews to expand database coverage, include multilingual research, and possibly incorporate preprints.

Findings

Following the PRISMA screening process described in the methodology, 32 studies were included in the SLR. These studies were analyzed to identify patterns in prompt injection attack strategies, attack success rates, vulnerable models, and proposed mitigation approaches. Table 2 provides an overview of the included studies categorized by their primary research focus.

Types of Prompt Injection Attacks (RQ1)

The analysis of the reviewed studies revealed that there is no universally accepted taxonomy of prompt injection attacks. However, one widely referenced classification distinguishes between Direct Prompt Injection (DPI) and Indirect Prompt Injection (IPI), as shown in Table 3. These categories represent the foundational classification most frequently cited throughout the literature.

Direct prompt injection occurs when an attacker inputs a natural-language malicious prompt into an LLM with the intent to alter its output or circumvent its security mechanisms. In

contrast, indirect prompt injection embeds malicious instructions into external sources of information retrieved by the LLM for processing. External sources include websites, documents, emails, and databases (Zhan et al., 2024),. These attacks can result in unintended actions from systems, such as in data exfiltration, response manipulation, and data poisoning.

Although this classification provides a baseline for understanding, it is no longer sufficient to describe the expanding types of prompt injections identified in recent research (Ramesh et al., 2025);(Yu et al., 2025). Therefore, to better represent the range of attack patterns described in the papers studied, a thematic analysis was conducted to identify common themes and characteristics in the literature (Braun & Clark, 2006). The thematic analysis identified four overarching themes that characterize prompt injection attacks: methods of attack construction, manipulative strategies, target attack surfaces, and the evolution of attacks, summarized in Table 4.

The first theme pertains to the construction of adversarial prompts. Studies distinguish between two construction processes: manually created prompts employing persona adoption and instruction overrides, and automatically generated prompts using techniques such as GCG, gradient-based optimization, AutoDAN, and LLM-Fuzzers, the last of which stands out for its evolutionary and mutation-based approaches (Xu et al., 2024);(Yu et al., 2025).

The second theme concerns adversarial strategies manipulating a model's output. These attacks exploit an LLM's tendency to follow instructions and maintain conversational coherence by manipulating context and obfuscating instructions. It also creates a narration that persuades LLMs to comply with malicious instructions (Zhang et al., 2024);(Ramesh et al., 2025);(Wang et al., 2024).

The third theme explores the attack surface. While previous attacks focused primarily on individual LLMs, recent research expanded its scope to LLM-based application platforms,

demonstrating practical experiments on translation engines, ranking engines, and database-connected agents (Pedro et al., 2025);(Pfrommer et al., 2024).

The final theme illustrates the trend of evolving prompt injection attacks constructed with emerging techniques such as automation, the extension of attacks to multi-turn interactions, and increased transferability of adversarial attacks across different LLMs (Ramesh et al., 2025);(Yu et al., 2025);(Lin et al., 2025).

Success Rates of Prompt Injection Attacks (RQ2)

The reviewed studies consistently demonstrate that prompt injection attacks are highly effective across different models and attack techniques. Attack performance is typically measured using Attack Success Rate (ASR), which represents the proportion of malicious prompts that successfully bypass safety mechanisms and produce restricted outputs (Kaul et al., 2025).

Several evaluation methods are used to determine ASR, including keyword-based refusal detection, external LLM-based judges, and human evaluation methods (Shayegani et al., 2023);(Andriushchenko et al., 2025);(Xu et al., 2024). Some studies also measure additional metrics such as computational cost, efficiency, and the impact of attacks on benign model performance.

Across the reviewed literature, prompt injection attacks frequently achieve ASR values exceeding 70%, with several techniques approaching near-perfect success rates. Template-based jailbreak prompts demonstrated particularly high effectiveness, with some prompt templates achieving 100% ASR against GPT-3.5-Turbo (Xu et al., 2024). Persona-based attacks such as DAN variants also achieved success rates exceeding 95% in multiple evaluations (Shen et al., 2023).

Optimization-based attack techniques demonstrate strong performance. Gradient-based approaches, such as GCG, achieved success rates of approximately 90%, while evolutionary algorithms, such as AutoDAN, achieved 76.9% ASR (Zhang et al., 2024);(Zhou et al., 2024). Iterative refinement techniques such as PAIR achieved approximately 80% ASR (Xu et al., 2024).

Psychological manipulation strategies also demonstrate high success rates. Context manipulation attacks achieved 100% ASR in some studies (Xu et al., 2024), whereas narrative persuasion techniques, such as Sequence of Context (SoC), achieved 92–95% ASR (Ramesh et al., 2025).

Domain-specific attacks can be particularly severe. Prompt-to-SQL (P2SQL) attacks successfully compromised database-integrated LLM applications with 100% ASR in controlled testing environments (Pedro et al., 2025).

Overall, the reviewed studies indicate that prompt injection attacks remain highly effective across models and deployment contexts, highlighting the seriousness of this vulnerability in real-world LLM systems.

Vulnerable LLM Models (RQ3)

The reviewed literature suggests that susceptibility to prompt injection is a vulnerability affecting both proprietary and open-source models across different architectures.

Among proprietary models, OpenAI GPT-3.5, GPT-4, Claude, & Google PaLM were frequently evaluated in literature, with GPT-3.5 demonstrating high vulnerability, often with attack success rates (ASR) of above 90% (Xu et al., 2024);(Shayegani et al., 2023); GPT-4 has also been successfully attacked, with some studies showing attack success rates of more than 95% under certain conditions (Sun et al., 2024).

Open-source models have shown similar levels of vulnerability. Success rates for attack on the LLaMA family of models, including LLaMA-3 variants, exceeded 85-95% (Pfrommer et al., 2024); fine-tuned models like Vicuna have shown extreme vulnerability, with many demonstrations showing almost 100% ASR (Xu et al., 2024). Attacks have also been successfully conducted on other models, including Mistral, Mixtral, ChatGLM & Dolly.

Multimodal models introduce an additional attack surface. Vision-language models, such as LLaMA, MiniGPT-4, and InstructBLIP, have been successfully attacked via prompt injections included in image inputs, with reported attack success rates of 50-84% (Xu et al., 2024);(Shayegani et al., 2023).

Additionally, several studies have demonstrated that prompt injection attacks can compromise LLM-integrated systems, including database-connected apps, search engines, and externally connected API apps. These attacks demonstrated successful data exfiltration, output manipulation, and unauthorized system actions (Pedro et al., 2025);(Pfrommer et al., 2024).

These findings suggest that prompt injection represents a systemic vulnerability affecting multiple model architectures and deployment environments rather than an issue limited to specific models.

Approaches to Defending Against Prompt Injection (RQ4)

The reviewed studies propose several categories of defensive approaches to mitigate prompt injection attacks. The analysis identified six major categories of defense mechanisms is summarized in Table 5.

The first category of defenses employs a series of model-alignment-based safeguards, including Reinforcement Learning from Human Feedback (RLHF) and vendor-provided safety patches, to reduce the likelihood of harmful outputs (Shen et al., 2023). Some systems extend this

approach by enabling models to evaluate their own responses before generation for potential safety violations.

The second category comprises external moderation filters that analyze the provided prompt or the LLM-generated response. Moderation systems such as OpenAI's Moderation API, WildGuard, LLaMA-Guard, and DataSentinel are prominent examples categorizing both prompts and generated responses based on safety (Liu et al., 2025);(Hu et al., 2024);(Han et al., 2024).

The third category contains prompt-level defensive mechanisms designed to prevent malicious prompts by altering the structure of the prompt itself. Such techniques include instruction delimitation, prompt paraphrasing, self-reminder prompts, and intentional analysis techniques detecting the user's intended level of harm by assessing their prompt (Zhang et al., 2024);(Kaul et al., 2025);(Zhang et al., 2024);(Lin et al., 2024).

The fourth category includes input modification techniques that alter or perturb a prompt before it reaches the model. Techniques such as SmoothLLM, perplexity-based filtering, and information bottleneck methods all disrupt adversarial patterns embedded within the prompts (Kaul et al., 2025);(Hu et al., 2024);(Liu et al., 2024).

The fifth category involves architectures designed to defend against adversaries while operating entirely within the model during inference. Architectures such as Hidden State Filtering, GradSafe, RobustKV, and SafeDecoding, are all examples of architectures that use internal model signals to detect adversarial inputs (Xie et al., 2024);(Qian et al., 2025);(Jiang et al., 2024);(Xu et al., 2024).

The final category includes multi-layered defense frameworks that combine multiple defensive strategies to provide coordinated protection for the system. Architectures such as ALIS, MART, GuidelineLLM, LangShield, and CIDER, are all examples of multi-layered defense

frameworks that employ defensive strategies throughout multiple layers of the LLM interaction pipeline (Song et al., 2025);(Zhang et al., 2025);(Ge et al., 2023);(Pedro et al., 2025).

Evidence of Defense Effectiveness (RQ5)

Built-in safeguards such as RLHF provide a baseline level of protection but remain vulnerable to semantic manipulation. Minor paraphrasing can restore attack success rates from 47% to 85%, demonstrating the fragility of alignment-based defenses (Shen et al., 2023).

Studies using external moderation systems have demonstrated promising success in detecting known forms of attacks. For instance, WildGuard significantly decreased ASR from 79.8% to 2.4%. The LLaMA-Guard model reported a classification F1-score of 81.9% (Xie et al., 2024);(Han et al., 2024). However, many systems struggle to identify new attack types or those that have been highly obfuscated (Lin et al., 2025).

Prompt-specific defenses have yielded inconsistent results. Intention analysis reduced ASR from 72.7% to 3.79% in controlled environments (Zhang et al., 2024). However, other prompt-specific defenses, including delimiter-based prompts, remain vulnerable to attacks, yielding ASR values above 78% (Zhang et al., 2024).

Input transformation and architectural defenses provide additional defense; however, they introduce trade-offs of increased computational costs and inference latencies. For example, paraphrasing-based defense resulted in increased inference times of approximately 80%, while some studies indicated that intention analysis-based models had inference times double those of baseline models (Zhang et al., 2025);(Xu et al., 2024).

Overall, the research indicates that current defenses provide incremental improvements over previous defenses but do not provide comprehensive protection against attacks. Additionally, evidence proves that current approaches remain vulnerable to adaptive adversarial attacks. These

findings indicate a need for more robust and systematic approaches to securing LLM-based applications.

Discussion

The findings of this SLR highlight the growing security risks posed by prompt injection attacks on LLMs, demonstrating that increasingly sophisticated attacks can successfully compromise models across different architectures and deployment environments.

One important observation from the reviewed literature is the increasing sophistication of prompt injection techniques. Early prompt injection attacks primarily relied on manually crafted prompts designed to bypass safety mechanisms through instruction overrides or role-playing prompts. However, recent studies demonstrate the emergence of automated attack generation techniques capable of producing adversarial prompts at scale. Optimization-based approaches such as gradient-based prompt generation and evolutionary mutation strategies allow attackers to systematically refine prompts until they successfully bypass model safeguards (Xu et al., 2024);(Yu et al., 2025). In addition, multi-turn conversational strategies such as Sequence of Context (SoC) attacks exploit conversational dynamics to gradually manipulate model behavior and increase attack effectiveness (Ramesh et al., 2025). These developments suggest that prompt injection attacks are evolving from simple prompt manipulation techniques into scalable adversarial methods capable of adapting to defensive mechanisms.

The review also revealed that all tested models are susceptible to prompt-injections, regardless of architecture or proprietary (Xu et al., 2024);(Pfrommer et al., 2024). This is evidenced by research on GPT-based systems, LLaMA-derived models, and other architectures (Xu et al., 2024);(Pfrommer et al., 2024). In multiple tests, models such as GPT-3.5 and fine-tuned LLaMA models achieved ASR exceeding 90% (Xu et al., 2024);(Shayegani et al., 2023).

Moreover, Multimodal models increase the attack surface as prompt injections are embedded within images and other forms of input (Xu et al., 2024);(Shayegani et al., 2023). This suggests that prompt injection is a systemic flaw in LLM design rather than an isolated weakness in specific implementations.

The review indicated numerous limitations in existing defenses against the prompt injections. Primarily, proposed defenses rely on model alignment, such as RLHF, or external moderation filters to prevent harmful outputs (Shen et al., 2023);(Liu et al., 2025). While some studies indicate their success, most show that adversarial prompts can circumvent them through semantic manipulation, contextual framing, and automated mutations of the prompt (Shen et al., 2023);(Zhang et al., 2024). As a result, many defenses provide partial mitigation rather than comprehensive protection.

The reviewed literature also a challenge for developing robust defenses. Defensive techniques such as prompt paraphrasing, adversarial filtering, and intention analysis are likely to increase latency and the computational resources required for model processing. This trade-off negatively affects the performance of the model (Zhang et al., 2025);(Xu et al., 2024). Additionally, these defensive mechanisms may affect legitimate queries given to the model, thereby reducing utility and response quality. The trade-offs described above emphasize the challenges in developing effective security mechanisms that provide strong defense against adversarial prompts while maintaining acceptable levels of model performance and usability.

This review has also demonstrated that most current defensive mechanisms adopt a model-centric approach to defending against prompt injection attacks, focusing on limiting the language model's ability to generate responses containing sensitive information. Other studies show that prompt injection attacks often exploit this vulnerability to compromise the larger systems

integrating LLMs, as opposed to solely being a risk to the model itself. Prompt-to-SQL attacks have demonstrated that maliciously crafted prompts can cause LLM-based database interfaces to issue SQL queries, potentially enabling unauthorized access to or modification of sensitive data (Pedro et al., 2025). Similar risks affect LLM applications interacting with third-party API's, search engines, and internal enterprise data systems.

This observation indicates that prompt injections should be viewed as more than a model-specific security concern; it is an architectural challenge affecting the interaction between LLMs and the external systems they control. As LLMs integrate into complex applications executing tasks, retrieving data, or interacting with operational systems, prompt injection attacks pose risks to output generation, system manipulation, data exfiltration, and unintended execution of actions.

Collectively, these findings underscore the need for additional security measures to address prompt injection-related vulnerabilities across the entire LLM application architecture. Future research in this domain would likely benefit from developing defensive strategies that extend beyond model-based security mechanisms and incorporate system-based security measures to protect interactions between LLMs and external systems.

The literature also identifies a need for standardized evaluation methodologies for prompt injection defenses. In many instances, researchers have employed different experimental designs, datasets, and evaluation metrics to assess the effectiveness of their proposed defense mechanisms. It is therefore difficult to directly compare the results of various studies. The development of standardized benchmarks and evaluation frameworks would provide a means to assess the reliability of various defense mechanisms and facilitate greater consistency in evaluations of defensive techniques in this area.

The results of this SLR suggest major implications for countries adopting artificial intelligence technology into their national digital infrastructures. With national strategies like the UAE Artificial Intelligence Strategy 2031 and the National Cybersecurity Strategy emphasizing the development of trustworthy, secure, and resilient AI systems, it is essential to protect the integrity of the AI systems and preserve the trust of users in AI enabled applications and protection of critical digital infrastructure and assets, given the increasing use of LLMs across all government services, enterprise systems, and digital platforms.

Overall, this literature review finds that prompt injection attacks will remain a potential threat to LLM security. It is therefore imperative to address the threat to ensure the safe and successful deployment of AI-powered systems in real-world applications.

Conclusion

This research conducted an SLR to assess prompt injection attacks against LLMs and the proposed defensive methods. The literature emphasized that prompt injection attacks represent a significant and evolving vulnerability affecting modern LLM systems. Furthermore, numerous studies demonstrated that adversarial prompts achieved very high attack success rates, indicating that existing safeguards remain insufficient to fully protect LLM-integrated applications.

In addition to underscoring the growing complexity of prompt injection attacks, the literature also highlights that current defense strategies are mostly model-centric, with most practical methods focusing on improving model alignment and filtering out prompts. While some current defense mechanisms can limit or even eliminate certain types of attacks, the literature indicates that many defenses remain vulnerable to adversarial adaptation.

Overall, this review highlights the need for more comprehensive approaches to securing LLM-based systems. As language models are increasingly integrated into real-world applications

and enterprise environments, addressing prompt injection vulnerabilities will be essential to ensuring the safe and reliable deployment of AI-powered technologies. Addressing these vulnerabilities will be particularly important for countries such as the United Arab Emirates that are actively advancing national AI adoption through initiatives such as the UAE Artificial Intelligence Strategy 2031.

Future research should therefore explore security strategies that extend beyond model-level safeguards and consider system-level protections capable of mitigating prompt injection attacks across the broader LLM application architecture.

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Tables

Table 1

Systematic Literature Review Inclusion and Exclusion Criteria

Screening Category	Inclusion Criteria (IC)	Exclusion Criteria (EC)
Language and Timeframe	IC1 – The article is in English.	EC1 – The article is earlier than 2021.
Relevance to Research Topic	IC2 – The article explicitly addresses prompt injection, jailbreaks, or related adversarial prompt attacks on LLMs.	EC2 – The article addresses unrelated AI security topics.
Evidence & Methodology	IC3 – The article is evidence-based.	EC3 – The article is conceptual or theoretical.
Quality & Publication Type	IC4 – The article is published in a conference paper or journal.	EC4 – The article is not peer-reviewed

Note: IC denotes inclusion criteria and EC denotes exclusion criteria used to determine the eligibility of studies during the systematic literature review screening process.

Table 2

Distribution of Studies Included in the Systematic Literature Review

Category	Number of Studies
Total studies included in the SLR	32
Studies analyzing prompt injection / jailbreak attacks	12
Studies proposing or evaluating defense mechanisms	1
Studies examining both attacks and defenses	18

Note: The table summarizes the number of studies included in the systematic literature review by research focus.

Table 3

Foundational Classification of Prompt Injection Attacks

Category	Definition	References
Direct Prompt Injection (DPI)	The attacker inputs the maliciously crafted prompt directly to the LLM agent or application to manipulate its output and bypass safety filters. This is the most straightforward form of attack.	(Zhang et al., 2024), (Zhan et al., 2024), (Wu et al., 2025)
Indirect Prompt Injection (IPI)	The malicious instruction is embedded within an external data source that the LLM retrieves and processes such as websites, emails, documents, or database entries, leading to data exfiltration, data poisoning, or answer manipulation without the user's knowledge.	(Zhang et al., 2024), (Zhan et al., 2024), (Wu et al., 2025), (Pedro et al., 2025), (Pfrommer et al., 2024)

Note: The categories represent the foundational classification of prompt injection attacks most frequently cited across the reviewed literature.

Table 4

Synthesized Themes of Prompt Injection Attacks Identified in the Literature

Theme	Description	Representative Attacks
Method of Attack Construction	Techniques used to generate adversarial prompts, including manually crafted prompts and automated optimization methods.	Persona prompts (DAN), instructional override, GCG, AutoDAN, LLM-Fuzzer
Manipulative Strategy & Psychological Leverage	Attacks that exploit conversational behavior and instruction-following tendencies of LLMs through persuasion, deception, or obfuscation.	Context manipulation, narrative persuasion (SoC), distraction attacks
Target & Attack Surface	Attacks targeting systems interacting with LLMs, including multimodal models and integrated applications such as databases or APIs.	Prompt-to-SQL (P2SQL), multimodal prompt injection
Evolution & Automation of Attacks	Increasing use of automated adversarial generation and multi-turn attack strategies that improve scalability and transferability.	Multi-turn attacks (SoC), automated prompt generation

Note: The themes were derived through thematic analysis of the reviewed literature on prompt injection attacks and represent the primary categories of attack techniques identified across studies.

Table 5

Categories of Prompt Injection Defense Mechanisms Identified in the Literature

Defense Category	Description	Representative Techniques
Built-in Safeguards	Alignment mechanisms integrated into the model during training to discourage harmful outputs.	RLHF, safety alignment
External Moderation Filters	Systems that analyze prompts or outputs outside the model to detect harmful content.	WildGuard, LLaMA-Guard
Prompt-Level Defenses	Techniques that modify or structure prompts to reduce susceptibility to malicious instructions.	Delimiters, self-reminders, intention analysis
Input Transformation	Methods that modify or perturb prompts before processing to disrupt adversarial patterns.	SmoothLLM, paraphrasing
Architectural Defenses	Internal model-level mechanisms that detect adversarial prompts during inference.	Hidden State Filtering, GradSafe
Multi-Layer Defense Frameworks	Integrated systems combining multiple defensive strategies across the model pipeline.	ALIS, MART, LangShield

Note: The defense categories represent the primary approaches proposed in the literature for mitigating prompt injection attacks against large language models.

Figures

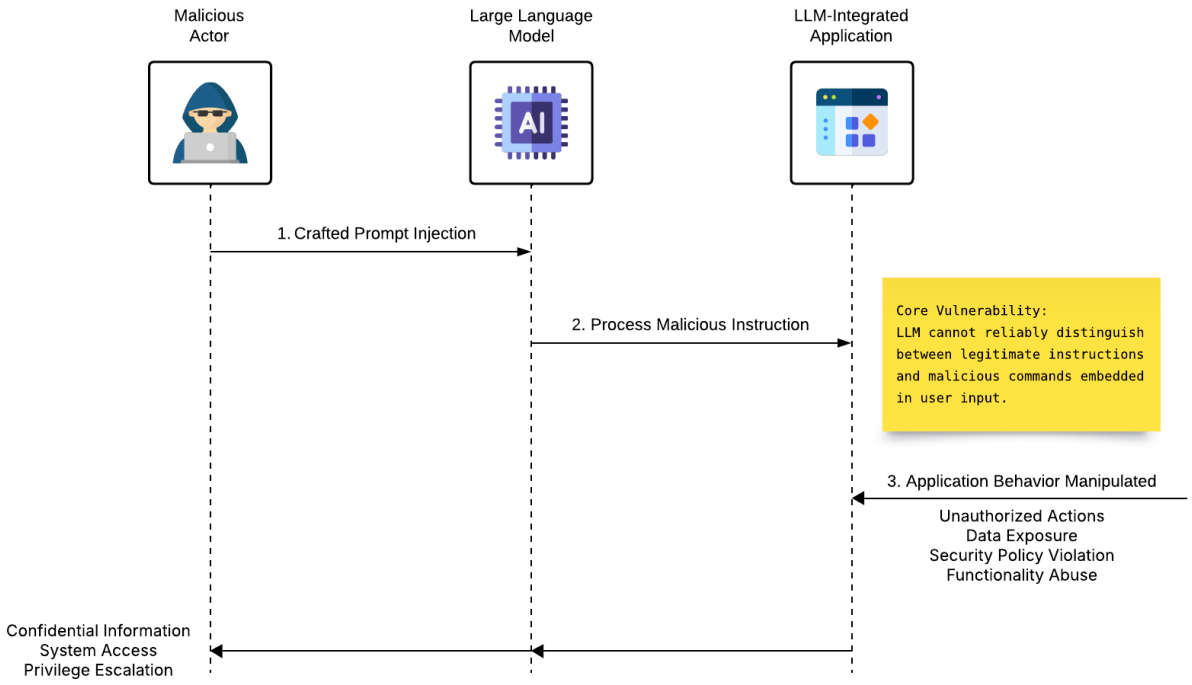

Figure 1. Prompt Injection Attack Flow - Diagram illustrating how malicious prompts manipulate LLM instructions and system interactions.

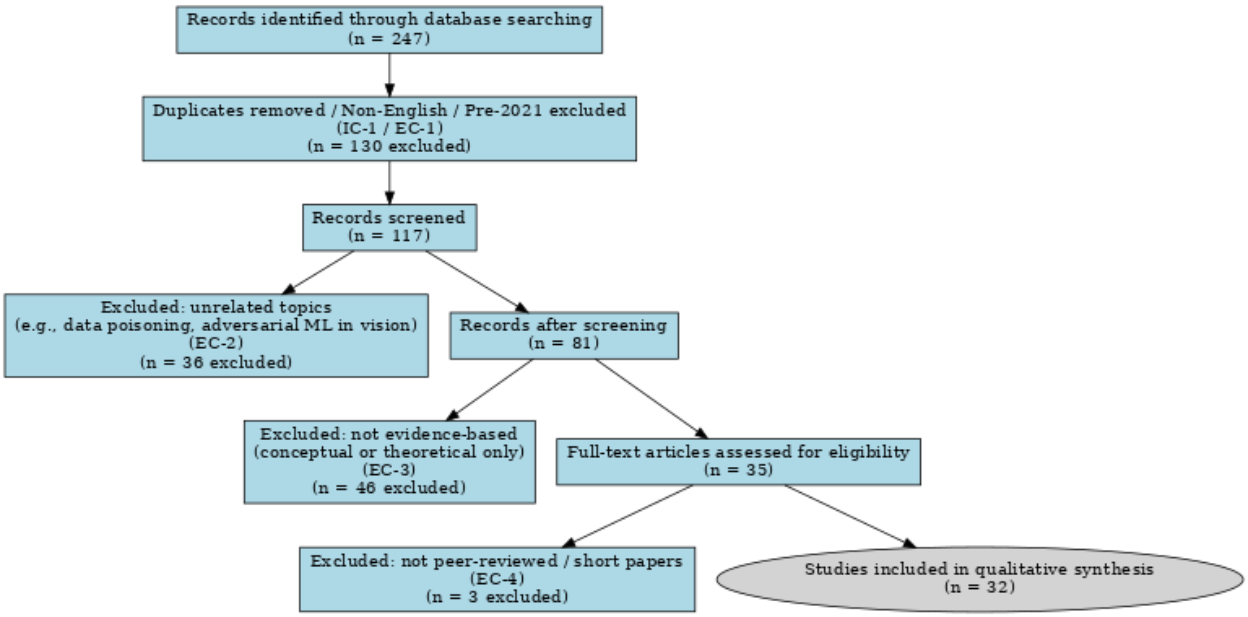

Figure 2. PRISMA Flow Diagram of Study Selection - Diagram illustrating the identification, screening, eligibility, and inclusion stages of the study selection process.

Appendices

Appendix A: Search Strategy

The literature search was conducted using the Scopus database. The following search query was used to identify studies related to prompt injection attacks in large language models.

Search Query (Scopus):

("prompt injection" OR "prompt attack" OR "prompt-based attack" OR
"prompt manipulation" OR "LLM jailbreak" OR "adversarial prompt" OR
"prompt hacking")

AND

("large language model" OR "LLM" OR "GPT" OR "generative AI")

Database: Scopus

Date of search: September 2025.

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